

Khansa Khanam

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Research Interests

My research examines how NLP and LLM systems encode and amplify bias in healthcare and socially sensitive domains, and how these failures can be measured for multilingual speakers, minority populations, and the resource-constrained organizations that serve them. I build evaluation frameworks that surface failure modes before deployment, and I am seeking a PhD in human-centered AI to pursue this agenda with rigor.

Education

Jan 2024 – Jun 2025 **M.S., Computer Science**, University at Buffalo (SUNY), Buffalo, NY
Aug 2017 – Jul 2021 **B.Tech., Information Technology**, Anna University, Chennai, India

Research Experience

Apr 2026–Present	Research Assistant (Part-Time) Rutgers University, School of Communication and Information, New Brunswick, NJ Advised by Prof. Tawfiq Ammari and Prof. Robert Wolfe - Investigating whether racial documentation bias in clinical notes (disparities in how patients are described, independent of medical content) is detectable by computational classifiers and human readers alike. - Tested race-predictive documentation patterns using open-weight LLMs (Llama-2-7B, Bode-7B-PT; retrieval-augmented Llama-3.1-8B), diagnosing context-window truncation as the dominant failure mode; a feature-engineered SGDClassifier (BERTopic/TF-IDF/LIWC) outperformed the LLM configurations tested (0.70 macro F1, cross-validated), but held-out testing surfaced a sharp per-class disparity (F1 = 0.72 white vs. 0.30 Black patients, $n = 27$) — an early, small-sample signal of unequal model legibility across groups. - Manually double-coded 123 notes against a testimonial-injustice and stigmatizing-language schema; found near-perfect human–human agreement on factual SDOH presence ($\kappa = 0.85$) but weak agreement on subjective bias markers ($\kappa \leq 0.43$), indicating automated bias detection did not substitute for expert coding in this setting. - Parallel workstream: evaluated Llama-3.3-70B on untranslated Brazilian Portuguese notes for SDOH extraction, validated against a two-annotator sample ($\kappa = 0.70$).
Jan 2026–Jun 2026	Research Contractor (Full-Time) University at Buffalo, Dept. of AI & Society, Buffalo, NY Advised by Prof. Kenneth Joseph and Prof. Joy Ming — work led to a CHI 2027 submission (see Publications) Contributed to inductive thematic analysis of design-workshop transcripts from caregivers, administrators, and developers at member-owned home care cooperatives, surfacing cross-cutting themes (resource tradeoffs, information privacy, worker voice) that informed the paper’s synthesis of benefits, harms, and design opportunities for AI in care-cooperative settings.
Jul 2025–Feb 2026	Independent Research (Volunteer) University at Buffalo, Dept. of Communication, Buffalo, NY Advised by Prof. Yotam Ophir - Studied how the public perceives and evaluates AI mental health tools in naturalistic use, and whether a hybrid human+LLM pipeline can reliably annotate this discourse at scale. - Designed and implemented the full technical pipeline independently: fine-tuned <code>mental/mental-roberta</code> with class weighting for severe label imbalance over $\sim 20,000$ Reddit posts (four-way functional-sentiment schema) and routed low-confidence predictions to an LLM for re-classification. - Validated the hybrid annotation approach against ~ 800 posts hand-labeled by a practicing psychiatrist (raw agreement, Cohen’s κ, Krippendorff’s α). Manuscript in preparation.

Sep 2024–Jan 2026

Research Assistant (Part-Time)

Institute of AI and Data Science, University at Buffalo, Buffalo, NY

Advised by Prof. Ashlee Ford Versypt

- Co-developed a constrained Bayesian optimization framework for calibrating agent-based (CompuCell3D) models of collective cancer cell invasion, with Gaussian Process surrogates and a feasibility-informed acquisition function, replacing manual heuristic tuning with a reproducible, uncertainty-aware search.
- Identified experimentally consistent parameter regimes for spheroid and network invasion phenotypes on a Pareto-optimal accuracy/feasibility trade-off. Manuscript in preparation; presented by co-authors at ECMTB 2026, SIAM LS26, and FOPAM 2026.

Publications & Manuscripts

Under review:

1. Ming, J., Tolera, H., Henriquez, O., Wurst, C., Khanam, K., Sultan, U., Hall, S., Joseph, K., & Gusoff, G. “Centering Care and Cooperation in the Design of AI Systems with Home Care Worker Cooperatives in the United States.” ACM CHI Conference on Human Factors in Computing Systems (CHI 2027).

In preparation:

- Benson, T. O., Krishna, A., Khanam, K., & Ford Versypt, A. N. “Bayesian Optimization Approach for Agent-Based Modeling of Cancer Cell Migration and Invasion Phenotype.”
- Multilingual evaluation of open-weight LLMs for Social Determinants of Health extraction from clinical notes (Rutgers SC&I).
- “AI perception among Reddit users seeking mental health support” (with G. Raffloer and colleagues, Dept. of Communication, University at Buffalo).

Industry Experience

Jun 2026–Present

AI Data Engineer, Manage Your Business, Buffalo, NY

- Architected end-to-end data infrastructure for an in-house ERP/CRM system on AWS, serving cross-functional teams across three countries.
- Built Airflow/dbt pipelines feeding downstream analytics, including ETL from unstructured sources into a real-estate underwriting model.

Technical Skills

Languages & Tools:

Python, SQL, AWS, Databricks, dbt, Airflow, Tableau, Git

ML / NLP:

Fine-tuning and evaluation of open-weight LLMs (Llama-2/3, Bode, Flan-T5), in-context learning, BERTopic, SGDClassifier, SHAP, prompt architecture design

Research Methods:

Mixed methods (qualitative + quantitative), participatory design studies, survey design, inter-annotator agreement (Cohen’s κ), Bayesian optimization, Gaussian Process surrogate modeling

Honors and Awards

2024

Grace Hopper Celebration Scholarship Recipient, AnitaB.org

2021

Finalist, AI21 Labs Hackathon